

Unveiling the Human Touch: Enhancing Customer Satisfaction Through Personal Profiles of Social Media Customer Service Agents

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Abstract

Firms are rapidly adopting social media platforms to optimize their customer service operations. Given the central role of social media as a primary point of contact with customers and the critical importance of maintaining high levels of customer satisfaction, it is imperative to craft positive experiences during service interactions. To enhance customer interaction with a personal touch, U.S. telecom giant T-Mobile replaced standardized profiles with personal profiles for its social media customer service agents on Twitter. Leveraging this exogenous change, we investigate how these personal profiles affect customer behaviors within service operations. We adopt a difference-in-differences empirical strategy in this research. The results show that implementing personal profiles for customer service agents increases positive sentiment in customer tweets, reduces the likelihood of complaints, and improves overall satisfaction. Further analyses suggest that the effects vary across customer groups, with greater impact among verified Twitter users and female customers. While personalized agent profiles can enhance satisfaction, they also heighten expectations for timely responses, making delays more detrimental than with standardized profiles. To explore the underlying mechanism, we conduct an online randomized experiment. In addition to reinforcing the causal inference from our secondary data analysis, causal mediation analysis of the experimental data shows that personal agent profiles influence satisfaction and the likelihood of customers expressing gratitude through the perceived warmth and perceived competence of the agents. Interestingly, while the effect on gratitude is mediated by humanization, we do not find a similar mediation effect on satisfaction. This suggests distinct pathways through which humanization shapes different aspects of the customer experience.

Keywords

Customer Service Operations, Social Media, Natural Experiment, Customer Satisfaction, Humanization

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I Introduction

Social technologies are digital products and services that enable and facilitate social interaction on online platforms. It allows people to engage in social interactions and collaboratively produce, contribute, and share content (Bughin et al., 2012; Chui et al., 2012). In recent years, social technologies, such as social media and social networks, have revolutionized the way customers and firms connect and engage with each other (Qiu and Whinston, 2017). Firms have embraced social technology for marketing, sales, communication, product research and development, production and operations, and customer service, etc. (Chui et al., 2012). Delivering customer service via social technology is among the most

rapidly expanding services offered by firms. According to Sprout Social, there was a 110% increase in the number of consumers who preferred to use social messaging to reach

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out to social media customer service between 2020 and 2021 (Lanier, 2022).

Social media customer service comprises two key stakeholders: firms' customer service operations and the customers themselves. Increasingly, firms are increasingly adopting social media as a dedicated service channel, thereby providing an alternative to traditional call centers (Chui et al., 2012). Many firms have embraced social media as a powerful customer service channel to actively monitor, handle inquiries and complaints, and prevent potential harm to businesses' reputations on social media platforms, such as Facebook and Twitter (now called X). Therefore, social media can be used as a service differentiator by firms looking to improve their customer service and create a competitive advantage (Das et al., 2022). For example, Guo et al. (2020) reports that the number of social media customer channels employed can positively affect firms' reputation scores. Moreover, customers are continuously switching to social media when they need customer support, as social media enables customers to reach out to firms and receive assistance in a more convenient and accessible way. Customers have gradually found that using social media to reach out for customer service is a more effective way of getting attention from brands, compared to contacting their customer service departments via telephone or email (Wolfe, 2018).

Social media customer service poses a unique challenge for firms due to its heightened visibility. Social media is one of the customer service channels with the most significant public exposure because the social media conversation and interaction between firms and customers is open and accessible to a wide audience (Das et al., 2022). Firms on social media must not only address their customers who are interacting with them on social media but also anyone who might be a witness to the public exchange. A high satisfaction rating from customers promotes the brand publicly, while, a low satisfaction rating harms the brand's image and undercuts brand loyalty (Bloemer and Kasper, 1995). As social media increasingly becomes the main point of customer contact and considering the importance of maintaining high levels of customer satisfaction on social media, it is crucial for firms to foster positive experiences during customer service interactions on social media.

One notable phenomenon commonly observed in the interaction between customers and service agents on social media is that customers often don't know whether they are talking to a real human agent or receiving an automated reply (a situation which has become more pronounced with the use of artificial intelligence [AI]). This confusion is often exacerbated when service agent profiles are standardized, typically featuring the company logo instead of the personal information of the service agent. As a result, the human touch of social media customer service operations has the potential to play an important role, as customers value the empathy and problem-solving abilities of human agents. Therefore, incorporating

humanized elements into service interactions will be valuable in the context of social media customer service.

It is worth noting that prior research has identified a relative lack of attention from operations management (OM) researchers regarding the field of customer service operations on social media (Guo et al., 2020). Only a limited number of studies have delved into service operations on social media platforms. For example, Gu and Ye (2014) examine the impact of online management responses on customer satisfaction. Fan and Niu (2016) explore the factors influencing the effectiveness of service recovery strategy on social media. Guo et al. (2020) investigate the relationship between social media-based customer service and firm reputation. While the existing literature primarily focuses on customer satisfaction and the impact of using social media for customer service on firm reputation, there is still a gap in understanding how firms' operational interfaces on social media platforms affect operational outcomes. More specifically, this involves examining how the operational interface with personal touch during service interactions on social media influences outcomes such as complaint volume and customer satisfaction levels. To address this gap, we examine the following research question in this paper. *What is the impact of service interface design, personal profiles compared with standardized profiles, of customer service agents on customer emotions and service satisfaction in social media customer service operations?*

To answer the research question, we conducted a field study that included collecting data from one of the largest social media platforms—Twitter.¹ On February 22, 2017, T-Mobile's official Twitter customer service account introduced a new feature designed to enhance the human touch. This involved adding personal profiles with personal information such as profile photos, names, personal interests, and biographies to their customer service agents, making it easier for customers or users² to identify and engage with service agents during their interactions. Taking advantage of this exogenous event for T-Mobile's customers who are also Twitter users, we investigate how agents' personal profiles affect customers' sentiments, the likelihood of tweeting complaints, and overall satisfaction, as expressed in their customer service tweets during interactions with customer service agents. We adopted a difference-in-differences (DID) specification and obtained tweets mentioning the official customer service Twitter accounts of four major U.S. telecom companies, AT&T Cares, Verizon Support, Sprint Care³ and T-Mobile Help, for three months before and after this event. Analysis of secondary data suggests that using personal profiles for social media customer service agents leads to increased positive sentiments, fewer complaint tweets, and higher customer satisfaction. We further test this effect through comprehensive online experiments, which confirm that personal profiles humanize the service interactions and enhance overall customer satisfaction in social media customer service operations.

The paper proceeds as follows. In Section 2, we review the literature. Section 3 discusses our hypotheses. Our empirical context, analysis, and findings are presented in Sections 4 through 6. Section 7 provides our conclusions as well as implications for research and practice and the study's limitations.

2 Related Literature

2.1 Social Media's Role in Enhancing Operational Performance

Social media, an advanced application of information systems, significantly enhances service operations by addressing OM challenges, such as improving operational performance (Kumar et al., 2018). The role of social media in OM and associated fields has been widely explored (Cheng et al., 2020; Huang et al., 2020; Lam et al., 2016). Research in this area has examined various impacts of social media, including its influence on new product development (Chan et al., 2016), firms' sentiment manipulation operations (Lee et al., 2017), team members' knowledge sharing and job performance (Cui et al., 2020), firms' operational efficiency and innovativeness (Lam et al., 2016), supply chain management (Orji et al., 2020), and service recovery (Fan and Niu, 2016). Recent studies have explored the dynamics of social media-based customer service from both provider and customer perspectives. Companies are more responsive to complaints from customers with larger followings, providing faster responses, yet are less likely to engage with complaints involving multiple parties, illustrating the bystander effect (Gunarathne et al., 2018). Regarding the communication method on social media customer service, although firms are moving towards private messaging for handling customer service, public complaints remain preferred by customers (He et al., 2024). However, not every social media post receives a response from customer service teams. Customers' politeness and high social status also increase the likelihood of receiving a company response (Hu et al., 2019, 2021). Previous research found that as the extent of social media service channels increases, firms can enhance their reputation by leveraging social media to absorb valuable knowledge from customer feedback and suggestions (Guo et al., 2020). Our study specifically addresses customer service operations in social media-based customer service, an area that has recently drawn increasing attention from researchers (Fan and Niu, 2016; Guo et al., 2020).

A closely related study by Gao et al. (2023) examines how identity cues influence customer engagement on social media customer service platforms. This study extends Gao et al. (2023) by going beyond the use of identity cues—such as agents using their first names instead of a two-letter code—to address the personalized service interface with a human touch. In summary, our study adds to this growing body of research by examining how changes in firms' social media customer service profiles (i.e., standard profile vs. personal profile)

affect customer behaviors, such as customer sentiments, complaints, and satisfaction. This study also echoes the call for exploring the firms' operations approach to address customer complaints on social media platforms (Gu and Ye, 2014; Guo et al., 2020).

2.2 Service Operations: E-services, Failures, and Design Strategies

Our study contributed to the literature on service operations, particularly to studies in these four areas: e-service operations, service failures, waiting time, and standardized vs customized service. The first focus of our study explores *e-service operations*. In an electronic retailer service setting, Heim and Sinha (2001) find ease of return, timeliness of delivery, and website navigation are the three most important factors contributing to customer loyalty. Field et al. (2004) develop a process model to assess and manage e-service quality by identifying the fundamental components of the e-service system and identifying sources of poor quality. Research highlights that e-service process attributes, derived from multisector datasets, significantly influence e-service quality dimensions, both positively and negatively, emphasizing the need for managers to prioritize these attributes in service process design (Heim and Field, 2007). Incorporating psychological constructs such as emotions, trust, and control into service design and delivery enhances positive customer perceptions, guiding managers to create effective e-service processes (Chase and Dasu, 2014). Our research contributes to this line of literature by explaining the importance of incorporating personal agent profiles in e-service design and delivery, enhancing customer service experience through a human touch and fostering emotional connection with agents.

The second area examines *service failures* and instances where firms' service performance falls short of customer expectations. Given the inevitability of such failures, it is vital for companies to manage these incidents effectively, develop recovery strategies, and learn from them (Lapr , 2011). For example, Craighead et al. (2004) investigated the influence of service failure severity and customer loyalty on recovery strategies. Lapr  (2011) explored the U-shaped learning-curve effect, highlighting that customer dissatisfaction often stems more from their propensity to complain to a third party about a service failure, rather than the failure itself. Mellat-Parast et al. (2015) demonstrated that customer complaints, a primary indicator of service failure, are a significant predictor of airline profitability. Our study explores the understudied role of service design, incorporating customized and personal agent profiles, in mitigating service failures and enhancing service performance.

The third stream investigated customer *waiting time* or customer service agents' response time. Customer waiting time can be viewed as a trade-off. For customers, an increase in the total waiting time signifies longer delays or greater congestion, which has a negative impact on firms. For service providers,

it suggests a higher rate of service system utilization (Chen et al., 2020). A study by Davis and Maggard (1990) on two-stage service process found that the initial wait before service begins (the “pre-process” wait) has a stronger negative impact on customer satisfaction compared to subsequent waits (the “in-process” wait) between service stages. Altman et al. (2020) indicated that when customers express more negative emotions in their messages, agents need to invest additional communication effort and extended response time in responding to resolve customer concerns or issues, ultimately extending the service conversation. Our study contributes to the literature on customer waiting time and agent response by examining how waiting time interacts with the presence of a personal agent profile to influence customer service satisfaction.

The fourth strand of research our paper addresses centers on the balance between *standardization and customization in service design*. Argote et al. (2003) argued that standardization, through its enhancement of knowledge transfer, offers a competitive edge in service OM. This strategy is commonly employed in various operational areas, such as service chain design. However, an over-reliance on uniformity can inhibit a service chain’s ability to adjust to local differences and achieve optimal performance. Kong et al. (2022) indicated that customized service delivery not only boosts financial outcomes but also positively impacts the welfare of residents in chain-affiliated nursing homes. Wang et al. (2010) explored the effects of standardized versus customized service offerings on customer satisfaction within the auto repair industry, finding that a mix of both can adversely affect satisfaction. Conversely, Kasiri et al. (2017) studied these dynamics across the hotel, healthcare, and education sectors, finding that a well-balanced approach is crucial for improving customer satisfaction and loyalty. Our research adds to these insights by investigating how standardized versus personalized service interfaces in social media customer service affect user behaviors and satisfaction. For a detailed comparison of our study with related research, please refer to the summary table in E-Companion A.

3 Hypothesis Development

We leverage the theoretical lens of social presence, which is a widely accepted theoretical framework for understanding subjects’ interaction in computer-mediated communication (CMC) (Bleier et al., 2019; Li and Xie, 2020; Ou et al., 2014; Pavlou et al., 2007). Social presence was initially defined as the extent to which another person is perceived as ‘real’ in an interaction, highlighting the importance of interpersonal relationships (Short et al., 1976). This theory posits that media with high social presence are perceived as sociable, warm, and personal, fostering a sense of human contact (Cyr et al., 2009; Short et al., 1976). It has also been conceptualized as the degree of feeling, perception, and reaction to another in CMC environments (Tu and McIsaac, 2002), later extending to include “the sense of being with another” or “the sense of

being together” in a virtual environment (Biocca et al., 2003). Environments or communication media with high social presence encourage individuals to recognize and respond to each other’s emotions, treating them as social beings rather than objects (Sia et al., 2002). Social presence is particularly crucial in CMC, where the digital format often diminishes face-to-face interaction, thereby reducing the sense of social presence and impacting the interaction’s humanness (Cobb, 2009; Pu et al., 2020; Sia et al., 2002).

In our context, we contend that a personal service agent profile, which includes the agent’s name, image, and bio, enhances social presence on social media platforms more effectively than a standardized profile displaying just the company logo. Social presence theory helps explain that technologies can exhibit different levels of human-likeness, or humanness based on factors such as the presence of social cues, anthropomorphism in virtual agents, and the type of technology and medium used (Lankton et al., 2015; Oh et al., 2018; Tsai et al., 2021). For example, social media technology like Facebook or Instagram are perceived as more human-like by users because they offer features such as users’ personal profiles, stories, or status update features to communicate with others (Lankton et al., 2015).

Websites featuring human face images are often perceived as conveying a higher degree of social presence compared to those without such images (Cyr et al., 2009; Hassanein and Head, 2007). In addition, a heightened sense of social presence can also be achieved by incorporating socially rich text, such as the personal bio of the service agent. In contrast, a standardized profile without an agent’s image and bio produces a low level of the agent’s social presence, promoting the feeling of an automatic algorithm reply from the social media platform or chatbot service agents. Sproull and Kiesler (1991) observed that the absence of social context cues, such as names, personal images, and bios, impairs the quality of interpersonal communication and leads to dehumanized perceptions of the communicating partner.

High social presence is perceived as sociable, warm, and personal, fostering a deeper sense of human contact during interactions (Cyr et al., 2009; Short et al., 1976). The personal service agent profile improves the intimacy and closeness of the interaction (Ou et al., 2014), and fosters positive emotions. By creating an environment where individuals feel more connected and understood, high social presence effectively bridges gaps in communication, making interactions more meaningful and emotionally fulfilling. We propose that a personal service agent profile promotes positive emotions of customers when seeking customer services on social media platforms. To capture the positive emotions of customers who seek customer services on social media, we use the positive sentiment in their posting and the likelihood of posting complaint tweets. This leads to the following hypotheses:

Hypothesis 1a (H1a): A personal service agent profile leads to more positive sentiment in customer service tweets on the social media platform.

Hypothesis 1b (H1b): A personal service agent profile reduces the likelihood of a customer service tweet being a complaint tweet on the social media platform.

Next, we argue that the high level of social presence from a personal service agent profile leads to higher customer satisfaction with social media customer services. First, the high level of social presence from a personal service agent profile may enhance customer satisfaction by improving the level of humanization of service agents on social media platforms. The personal profile of customer agents on a social media platform makes customers feel they are interacting with a human rather than a machine or algorithm. This perception significantly improves the intimacy and closeness of the interaction (Ou et al., 2014), and reduces the perceived social distance between the parties involved (Pavlou et al., 2007). Research shows that the degree of human-likeness in service interactions profoundly affects how customers perceive and evaluate their experiences. Studies have shown that human-like attributes of virtual agents or chatbots enhance perceptions of social presence, leading to higher customer satisfaction (Tsai et al., 2021; Verhagen et al., 2014), positively influence online purchase (Bleier et al., 2019) and higher user engagement on social media (Li and Xie, 2020). In contrast, a standardized profile featuring a company logo instead of an agent's image and bio promotes a sense of nonhumanness, robotic nature, and unsociability. This gives customers the feeling of human-robot interaction on social media platforms, which results in negative feelings regarding the interaction and service outcomes.

Second, a personal service agent profile on social media platforms enhances social presence, significantly boosting perceptions of the agent's human warmth and competence. Customers seeking service support often perceive humans are more friendly and capable to understand and resolve their issues. They become frustrated when the machine algorithm or chatbot doesn't understand their problem (CGS, 2018). The personal profile of customer agents on the social media platform makes customers feel like they are interacting with a human, which is more friendly and skillful than a machine or algorithm. These perceptions are critical in social judgment and cognition, as warmth includes traits like friendliness and helpfulness, while competence involves skill and efficacy (Fiske et al., 2007). High social presence can elicit positive customer perceptions, leading to better service outcomes (van Doorn et al., 2016). Agents perceived as warm and competent trigger positive emotions and behaviors in customers, contrasting with the negative reactions to those perceived as lacking these qualities (Fiske et al., 2007). Research shows that perceived warmth and competence guide consumer judgments and behaviors, influencing satisfaction, the likelihood of recommendations, overall enjoyment and positive attitude toward a website, and cooperative intentions during service recovery (Güntürkün et al., 2020; Scott et al., 2013). In our context, perceived warmth may make customers feel that the agent is friendly and helpful, while perceived competence may boost customers' confidence in the agent's ability to resolve their

issues, leading to higher customer satisfaction. Taken together, we propose:

Hypothesis 2 (H2): A personal service agent profile leads to higher customer satisfaction when customers engage with customer service agents on a social media platform.

4 Quasi-Experimental Setting and Data Description

4.1 Background

On February 22, 2017, Twitter introduced a novel customer service feature for Twitter Business accounts, enabling customer service agents to create a customized profile for their Twitter Business account (referred to hereinafter as a "profile") featuring personal identity information when interacting with customers. This service aimed to provide a more humanized customer service experience. T-Mobile, one of the largest mobile communications companies in the United States, was among the first companies, and the only telecom carrier in the United States, to adopt the new feature for its official customer service account, T-Mobile Help. It did so on the day the customized Twitter profile feature for business accounts became available.

Before the implementation of new personal profile features, customers interacting with T-Mobile Help on Twitter could only see the company logo and the initials or name of the customer service agent, as illustrated in Figure 1(a). This standard use of a brand logo might impart a mechanized impression, reducing the human element and fostering a sense of impersonality associated with automated services. With the adoption of the new profile features, T-Mobile Help's tweets began to include personal identity information, such as an agent's profile picture, full name, and a brief biography (shown in Figure 1(b)), allowing customers to see the personal profile of the engaging customer service agent. This change helped customers feel they were interacting with an authentic individual (Perez, 2017).

4.2 Natural Experiment Setting and Data

T-Mobile Help's adoption of the personal profile provided a unique opportunity to examine their causal effect on customer behaviors and satisfaction within the realm of social media customer service operations. The implementation of customized Twitter Business accounts, which became mandatory for all T-Mobile Help service agents, serves as an exogenous event for Twitter users and service agents. We leveraged this natural experiment to conduct our study, employing a DID estimation method (Bertrand et al., 2004; Lechner, 2010) to assess how personal profiles impact metrics like customer complaint and satisfaction.

T-Mobile Help's adoption of personal profiles for its customer service account uniquely positioned it as the sole member of the treatment group. To evaluate the impact of these

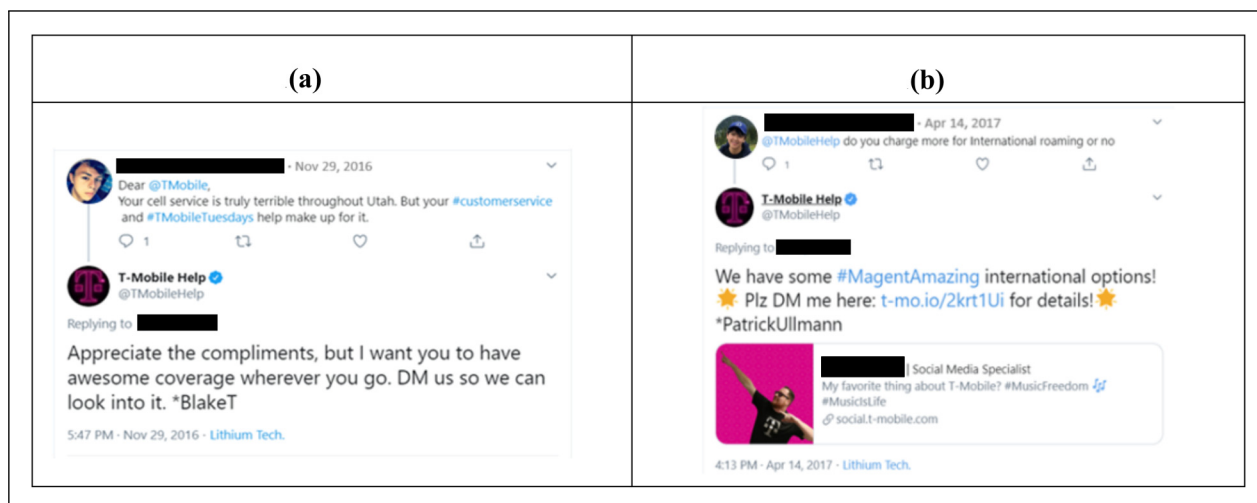


Figure 1. Customer service tweets from service agent before and after. (a) Preimplementation of personal profiles, (b) postimplementation of personal profiles.



Figure 2. Customer service tweet of control companies. (a) Sprint, (b) Verizon, (c) AT&T.

personal profiles on social media customer behaviors, we identified a relevant control group for baseline comparison. Our selection process for this control group was meticulous, as we opted for AT&T, Sprint, and Verizon. These major U.S. wireless carriers (2011–2019, Statista, 2020) did not use personal profiles during our study period, instead using standardized brand logos and signing tweets with names or initials, as shown in Figure 2(a)–(c). This distinction was crucial for assessing the impact of personal profiles on social media customer behaviors.⁴

Within this framework, we gathered tweets mentioning these four major U.S. telecom companies: AT&T Cares, Verizon Support, Sprint Care, and T-Mobile Help. Our data collection spanned from November 1, 2016, to May 31, 2017, with a specific emphasis on tweets occurring three months before and after T-Mobile Help introduced its personal profile on February 22, 2017. Additionally, the tweets we collected were consolidated into distinct categories of dialogs. We excluded duplicated tweets and telecom carrier-initiated dialogs, which were more likely to be advertisements, and focused instead

on customer-initiated dialogs, potentially capturing emotional responses from customers. Overall, the data set contained 100,555 user-initiated dialogs. We also collected timestamps, tweet content, tweet conversation threads, and Twitter users' profile information, such as the person's number of followers and number of accounts followed.

4.3 Measures

The main outcome variables are measures of customers' sentiments, complaints, and satisfaction. To ascertain the customers' sentiments in customer service tweets, we use a text analysis application called Linguistic Inquiry and Word Count (LIWC), which was developed in 2015. LIWC is a commonly used computer-based text analysis tool for exploring the psychological meaning of words. It can accurately identify emotions in language use (Tausczik and Pennebaker, 2009). It has also been applied to many works in the OM literature related to sentiment analysis (Wang et al., 2022). We employ "emotional tone" to measure the sentiment score in

customer service tweets, following the approach described by Wang et al. (2022). A higher sentiment score reflects a more positive attitude in the text, while a lower score shows more anxiety, sadness, or the expression of an unfriendly attitude (Pennebaker et al., 2015).

Second, we constructed a complaint measure based on the recent machine learning techniques for natural language processing (NLP)—Robustly Optimized Bidirectional Encoder Representations from Transformer Pretraining Approach (RoBERTa). Bidirectional Encoder Representations from Transformers (BERT) is widely considered as a breakthrough in the use of Machine Learning for NLP due to its creative way of understanding the meaning of ambiguous language in the text by using surrounding text to establish context (Devlin et al., 2018). RoBERTa modified the pretraining procedure of BERT with several improvements, including dynamically changing the masking pattern applied to the training data, removing BERT's next sentence prediction objective, and training the model with larger mini-batches, over a longer time and with more data (Liu et al., 2019).⁵ Here, our complaint variable is constructed as follows. We first classified user-initiated tweets separately from all other types of tweets. Second, 1500 tweets were randomly selected from the data set. Two authors worked independently to evaluate these tweets as a complaint or noncomplaint. If a tweet referred to a service or a product inquiry, and there were no negative terms used in it, it was categorized as a noncomplaint tweet. If a tweet expressed dissatisfaction toward a product or service, it was categorized as a complaint tweet.⁶ When there was a disagreement between the coders, we sought a third person's opinion and used the majority vote to break the tie. Third, we adopted RoBERTa to train the classifier with 1,500 manually classified tweets as the training dataset. The accuracy of the RoBERTa classifier was 87.33%,⁷ which was the highest among all the approaches, including original BERT and other conventional machine learning NLP methods. In addition, the detailed technical discussion on BERT vs RoBERTa, benchmark across NLP models, and tests with alternative measures of complaints are included in E-Companion D.

Third, for the satisfaction measure, we adopted Hu et al. (2021)'s approach to constructing the measure by identifying if the user expresses satisfaction in the conversation thread. Examples include statements like “Thank you for your speedy reply we will do that,” and “All set now. I spoke with a phone rep already. Thank you,” made after receiving responses from customer service agents. Additionally, users clicking on the “like” (Twitter's built-in feature) in response to agents' messages in the conversation thread are also considered indicators of satisfaction.

Further, we defined a dummy variable, *Post*, coded as 1 if the date was on or after the personal profile feature was adopted and 0 otherwise. *Treat* was a dummy variable coded as 1 if the telecom carrier was T-Mobile Help and coded as 0 if the carrier was AT&T Cares, Verizon Support, or Sprint Care. Our main interest is the interaction terms of *Post* and *Treat*,

since they capture the treatment effect of the agent's personal profiles on the social media platform.

Finally, we have several control variables at the user level and tweet level. Control variables of the user level are the public profile and tweeting behavior information of each Twitter user, such as the number of tweets posted, followers, likes, people the user followed, and user registration days on Twitter until the creation of the tweet, whether user profile contains description or location, and whether the user has the blue verified badge that is an account of public interest verified by Twitter. Control variables of tweet level are the characteristics of each user-initiated tweet and its conversation thread, such as the word counts of the user-initiated tweet, the length of Tweet conversation threads, customer service agent's sentiment in customer service tweets, and response time of customer service agents to the user's tweet. E-Companion B reports key information about our variables: Table A1a provides variable definitions, Table A1b presents summary statistics, and Table A1c displays the correlation matrix for the main variables.

5 Evidence From Quasi-Experimental Analyses

5.1 Main Results and Hypothesis Testing

We chose DID as our main empirical strategy because this approach is typically used to estimate the causal effect of specific treatment interventions and is a popular research design for estimating causal relationships in empirical economics and social science research (Bertrand et al., 2004; Lechner, 2010). We aim to examine the impact of a personal profile that includes a personal identity on customers' sentiments, complaints, and satisfaction on the social media platform. Our data include tweets from three months before and after T-Mobile Help adopted the personal profile. This specification incorporates telecom carrier fixed effects to control effectively for unobserved heterogeneity at the telecom companies. Following Kim et al. (2017), we also include month and day-of-week (DW) fixed effects. More specifically, the equation for the DID estimation is as follows:

$$y_{jit} = \beta_0 + \beta_1 Post_t + \beta_2 (Post_t * Treat_i) + \beta_3 Controls + Telecom\ FE + DW\ FE + Mon\ FE + u_{it} \quad (1)$$

where y_{jit} are sentiment, complaint, or satisfaction of a particular customer service tweet j to telecom carrier i at time t . $Treat_i$ is a dummy variable equal to 1 if the telecom carrier is T-Mobile Help, and 0 otherwise. $Post_t$ is a dummy variable indicating the postintervention period. We also include user-level controls such as profile, verified status, log number of total tweets, followers, following, likes, days registered; and tweet-level factors: log of tweet wordcounts and conversation length, the conversation thread ends with a user or customer service agent. Telecom carrier-specific fixed effect and DW as well as month (Mon) fixed effects are included. u_{it} is the

idiosyncratic error term. Standard errors are robust and clustered at the telecom carrier level. Our main interest is the coefficient of the interaction term β_2 , which captures the influence of a personal profile of social media customer service agents on the behavior of users on Twitter.

The results of the estimates for *sentiment* are presented in Table 1, Column (1). There, we see the coefficient of the interaction term *Post*Treat* is positive and significant, indicating customers are more likely to express positive emotions in customer service tweets after T-Mobile Help adopted the personal profiles for customer service agents. In particular, the personal profile adoption increases the sentiment score by 1.996 (4.8% of the average sentiment score of 41.5), which is arguably notable. Next, to assess the impact of personal agent profiles on customer complaints and satisfaction, we employ a linear probability model to estimate equation (1), as the dependent variables, complaint, and satisfaction, are binary.⁸ The estimation results for the likelihood of complaint are reported in Table 1, Column (2). The coefficient of *Post*Treat* is negative and statistically significant, indicating that personal profiles contribute to a reduction in the likelihood of a tweet being a complaint. Economically, the predicted probability of a customer service tweet being a complaint decreased by 1.1% after the adoption of personal profiles. This translates to a decrease of 7–8 complaint tweets per week, considering the average total number of weekly tweets across four platforms in our sample is 676.39. This is nontrivial given the severe consequences of negative word-of-mouth on social media. Table 1, Column (3) shows the estimated results of satisfaction. The coefficient of *Post*Treat* is positive and statistically significant. Specifically, the personal profile adoption increases the likelihood of expressed satisfaction by 3.4%, which translates to an increase of 23 tweets expressing “thank you” or clicking “like” given the average number of total tweets in a week of our sample is 676.39 (over four platforms). This is particularly significant considering the speed of information propagation on social media platforms. Overall, our evidence shows that personal profiles used by customer service agents in customer service tweets lead to an increase in the sentiments of customer service tweets (H1a), lower probability of being a complaint customer service tweet (H1b), and higher customer satisfaction (H2), providing support to our hypotheses.

5.2 Addressing Alternative Explanations

5.2.1 Alternative Control Group. As the baseline, the control group is composed of AT&T Cares, Verizon Support, and Sprint Care. A legitimate concern here is that a market leader in the telecom industry may adopt completely different market or customer care strategies from the treated telecom carrier T-Mobile, but unobserved by researchers. To address this concern, we exclude the market leader, AT&T, from our control group and estimate equation (1) with the alternative control group with only Verizon and Sprint.⁹ Table 2 columns (1), (2),

Table 1. Impact of agents' personal profiles.

	(1) Sentiment	(2) Complaint	(3) Satisfaction ¹⁵
<i>Post</i>	−1.7* (0.608)	0.014 (0.011)	−0.007 (0.008)
<i>Post*Treat</i>	1.996*** (0.267)	−0.011** (0.002)	0.034*** (0.003)
log (Total tweets)	−0.841*** (0.051)	−0.008 (0.004)	−0.011*** (0.001)
log (Followers)	0.783** (0.155)	−0.004 (0.004)	0.015*** (0.002)
log (Following)	0.106 (0.276)	−0.001 (0.003)	0.001 (0.002)
log (Likes)	0.127* (0.052)	0.002 (0.004)	0.015*** (0.002)
log (Word counts)	2.851** (0.561)	0.254*** (0.005)	−0.001 (0.007)
log (Register days)	0.085 (0.154)	0.006** (0.002)	−0.009** (0.002)
log (Conversation length)	−2.219*** (0.358)	0.033** (0.007)	0.269*** (0.008)
Profile	−0.173 (0.772)	0.002 (0.004)	0.002 (0.005)
Verified	3.244 (2.74)	−0.05 (0.056)	−0.036* (0.013)
Last	2.819* (1.127)	0 (0.019)	0.102** (0.021)
Telecoms FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes
Day of week FE	Yes	Yes	Yes
Obs.	100,555	100,555	59,590
R-squared	0.006	0.111	0.177

Robust standard errors are in parenthesis. *** $p < .01$, ** $p < .05$, * $p < .1$.

and (3) present the estimation results of the alternative control group. We observe that the coefficients of the interaction term *Post*Treat* are similar to our baseline model, and the results remain consistent. The results rule out the possibility that our findings are driven by differences between the market leader and followers, thus providing additional evidence for our findings that personal agent profiles lead to an increase in the sentiments of customer service tweets, lower chance of being a complaint tweet, and higher user satisfaction.

5.2.2 Personal Profile and Customer Service Agents. One alternative explanation for the observed effects is that customer service agents perform better after introducing the personal profile since their profiles are publicly disclosed (Forman et al., 2008; Pu et al., 2020), therefore, the enhanced performance of customer service agents explains the high sentiment, high satisfaction, and low complaints from T-Mobile Help customers after treatment. In order to address this alternative explanation, we conduct a test to examine the effects of personal profiles on agents' behaviors directly. More specifically, we examine the following aspects of the agent behavior and the tweet

Table 2. Addressing alternative explanation.

	Panel A: Control group only including verizon and sprint			Panel B: Agents' reaction to personal profiles		
	(1) Sentiment (w/o ATT) ¹⁶	(2) Complaint (w/o ATT)	(3) Satisfaction (w/o ATT)	(4) Agent's sentiment	(5) Response time	(6) Conversation length
Post	−0.917* (0.292)	0.016 (0.016)	−0.013 (0.009)	1.603 (0.82)	−663.994 (2326.51)	−0.021 (0.211)
Post*Treat	1.547*** (0.049)	−0.014* (0.004)	0.035** (0.005)	−0.25 (1.451)	−911.919 (1084.85)	−0.482 (0.262)
Tweet level controls	Yes	Yes	Yes	Yes	Yes	Yes
User level controls	Yes	Yes	Yes	Yes	Yes	Yes
Telecoms FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Day of week FE	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	63,790	63,790	41,628	59,590	59,590	59,590
R-squared	0.008	0.105	0.184	0.005	0.005	0.05

Robust standard errors are in parenthesis *** $p < .01$, ** $p < .05$, * $p < .1$.

conversations: (1) Agents' sentiment, measured by the sentiment score of customer service agents' responses; (2) agents' response time, defined as the duration between a customer's tweet and the agent's reply; and (3) measured by the number of exchanges in each tweet thread. If agents' behaviors were indeed affected, we would expect to observe a significant impact from the introduction of personal profiles.

Next, we re-estimated equation (1) with agents' behaviors as outcome variables and the results are reported in Table 2, columns (4)–(6). We observe that the coefficients of the interaction term *Post*Treat* are statistically insignificant across all three outcomes, which indicates that there are no significant changes in customer service agents' behaviors before and after the personal profile is introduced. These results first eliminate the possibility that our findings are caused by behavioral changes from customer service agents. Second, they cast doubt on the likelihood that operational improvements or technology upgrades explain our results, as such changes would likely have affected both customers and service agents.

Furthermore, we have conducted several robustness tests to reinforce the validity of our primary findings, and the detailed results are documented in E-Companion E. Our robustness testing begins with an examination of the parallel trend assumption to ensure the internal validity of our DID model. Subsequently, we investigate the resilience of our results by employing alternative measures of sentiment and negative emotions, addressing potential concerns about the reliance on our chosen measures. As a third step, we implement DID with propensity score matching (PSM) to ascertain the comparability of our treatment and control groups. Additionally, we consider the inclusion of various time-fixed effects to mitigate potential biases stemming from unobservable factors that may change over time. Importantly, all empirical evidence lends strong support to the main findings presented in Table 1.

5.3 Empirical Extensions

5.3.1 Verified vs. Unverified Users. Twitter supports account verification, signaling the legitimacy of well-known public figures such as actors, musicians, athletes, reporters, politicians, and entrepreneurs. The blue verified badge, awarded by Twitter officials, signifies the authenticity of an account of public interest (Twitter, 2019). Users can verify an account's authenticity by checking for the blue badge next to the account name. Verification enhances an account's credibility and fosters increased follower counts, as it helps users identify trusted Twitter accounts of public figures.

We conjecture that verified Twitter users differ significantly from unverified users and accordingly we examine heterogeneous treatment effects. Public figures and celebrities are inclined to share positive tweets for two main reasons. First, they must maintain their public image and enhance their reputations on social media platforms. Second, positive sentiment in tweets is known to attract more followers on Twitter (Hutto et al., 2013). Public figures aim to expand their social media influence by gaining more followers. If they find interacting with customer service agents with personal profiles to be a positive experience, they are likely to respond favorably and publicly express their positive experiences on Twitter. Collectively, we anticipate that the impact of personal profiles is stronger among verified Twitter users.

Next, we estimate the following regression equation to examine whether the impact of the personal profile varies based on a Twitter user's verification status:

$$\begin{aligned}
 y_{ijt} = & \beta_0 + \beta_1 Post_t + \beta_2 Verified_j + \beta_3 (Post_t * Verified_j) \\
 & + \beta_4 (Post_t * Treat_i) + \beta_5 (Treat_i * Verified_j) \\
 & + \beta_6 (Post_t * Treat_i * Verified_j) \\
 & + \beta_7 Controls + Telecom FE + DW FE \\
 & + Mon FE + u_{it}
 \end{aligned} \tag{2}$$

Table 3. Empirical extension—heterogeneous treatment effects.

	(1) Sentiment	(2) Complaint	(3) Satisfaction	(4) Sentiment	(5) Complaint	(6) Satisfaction
<i>Moderating variable</i>		Verified user			Female ¹⁷	
Post	−1.704* (0.602)	0.014 (0.011)	−0.007 (0.007)	−2.011*** (0.036)	0.02 (0.013)	−0.015 (0.016)
Moderating variable	1.308 (2.113)	−0.007 (0.018)	−0.035** (0.009)	−0.945 (0.585)	0.05*** (0.005)	0.015** (0.003)
Post*moderating variable	0.737 (2.393)	−0.006 (0.007)	0.008 (0.023)	−0.708* (0.261)	0.008* (0.003)	0.007 (0.007)
Post*treat	2.004*** (0.249)	−0.008** (0.002)	0.034*** (0.003)	1.272** (0.276)	−0.012** (0.003)	0.043** (0.01)
Treat*moderating variable	8.825** (1.838)	−0.092*** (0.01)	−0.065** (0.017)	1.773* (0.577)	−0.009 (0.004)	0.004 (0.004)
Post*treat	−4.61⁺	−0.10***	0.083**	0.916**	−0.007*	−0.014⁺
*Moderating variable	(2.304)	(0.004)	(0.024)	(0.233)	(0.003)	(0.007)
Tweet level controls	Yes	Yes	Yes	Yes	Yes	Yes
User level controls	Yes	Yes	Yes	Yes	Yes	Yes
Telecoms FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Day of week FE	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	100,555	100,555	59,590	47,602	47,602	28,823
R-squared	0.006	0.111	0.177	0.008	0.114	0.172

Robust standard errors are in parenthesis *** $p < .01$, ** $p < .05$, * $p < .1$, + $p < .1$ in one-tail test.

where y_{ijt} represents complaint, sentiment, or satisfaction. $Verified_j$ is a binary variable equal to 1 if a tweet is posted by a verified user account or 0 otherwise. The control variables are the same as those used in the DID model in equation (1). Our main focus is on the coefficient of the three-way interaction term, β_6 , which captures the impact of personal profiles on verified users compared to unverified users. The results are presented in Table 3 Columns (1)–(3). We observe that the verified user experienced a greater reduction in the likelihood of complaint (Table 3 column (2), $\beta_6 = -0.10$, $p = .01$) and were more likely to express gratitude either through responses or by clicking “Like” (Table 3 column (3), $\beta_6 = 0.083$, $p = .04$). These results align with our expectation that personal profiles have a significantly greater impact on verified users, who are often public figures or celebrities, compared to unverified users.¹⁰

5.3.2 Gender Difference. Previous studies have suggested that gender differences exist in expectations and perceptions in the context of CMC (Matheson, 1991). We hypothesize that the introduction of personal agent profiles would have an uneven impact on customers’ sentiments and complaints across genders. Specifically, we anticipate a larger effect on females than on males, considering that females tend to be more emotionally expressive (Kring and Gordon, 1998). To test this conjecture, we assess the heterogeneous treatment effects on both male and female Twitter users. We categorized users in our dataset into distinct gender groups, including males, females, and others (gender nonidentifiable),

using facial recognition techniques.¹¹ Our dataset contained 27,938 male users, 19,664 female users, and 52,953 users with an unidentifiable gender. For a more focused examination of gender-based differences, we excluded users with an unidentifiable gender from this analysis. Next, we estimated the following regression equation to investigate whether the implementation of personal service agent profiles has differing effects across genders.

$$\begin{aligned}
 y_{ijt} = & \beta_0 + \beta_1 Post_t + \beta_2 Female_j + \beta_3 (Post_t * Female_j) \\
 & + \beta_4 (Post_t * Treat_i) + \beta_5 (Treat_i * Female_j) \\
 & + \beta_6 (Post_t * Treat_i * Female_j) + \beta_7 Controls \\
 & + Telecom\ FE + DW\ FE + Mon\ FE + u_{it}
 \end{aligned} \quad (3)$$

where y_{ijt} refers to complaint, sentiment, or satisfaction. $Female_j$ is a binary variable equal to 1 if a tweet was posted by a female user or 0 otherwise. The variable of interest is the coefficient on the three-way interaction term, β_6 , which captures the impact of personal profiles on female users compared to male users. Control variables are the same as those in equation (1). The heterogeneous effects on gender are reported in Table 3 Columns (4)–(6). The results indicate that females experience a stronger increase in tweet sentiment (Table 3 column (4), $\beta_6 = 0.916$, $p = .03$) and are less likely to complain (Table 3 column (5), $\beta_6 = -0.007$, $p = .09$). In summary, the results are consistent with our expectation that the personal service agent profiles have a significantly larger effect on female users compared to male users.

Table 4. Empirical extension—wait time (agent's response time) and personal agent profile.

	(1) Satisfaction	(2) Satisfaction
Profile	0.1072*** (0.0073)	0.1501*** (0.0159)
Log (Wait time)	−0.0047*** (0.001)	−0.0028** (0.0012)
Profile*Log(Wait time)		−0.007*** (0.0023)
Controls	Yes	Yes
Telecoms FE	Yes	Yes
Month FE	Yes	Yes
Day of week FE	Yes	Yes
Observations	59,576	59,576
R-squared	0.1679	0.168

Robust standard errors are in parenthesis *** $p < .01$, ** $p < .05$, * $p < .1$.

5.3.3 Wait Time Tolerance. One important factor in service operations is wait time (Altman et al., 2020; Davis and Maggard, 1990). Customers often experience frustration when their issues are not addressed by an agent in a timely manner. In this analysis, we examine the interaction effect between personal agent profiles and wait time tolerance. Wait time is measured as the duration between a customer's initial service-related tweet and the agent's first response. In other words, it captures how long a customer waits for the agent's initial reply on the social media platform. Table 4 presents the results of two regression models assessing the effects of having a personal profile and wait time on customer satisfaction. In both models, having a personal profile is positively associated with satisfaction, with the coefficient increasing from 0.1072 in column (1) to 0.1501 in column (2) when the interaction term is included, consistent with our main findings. Log-transformed wait time is negatively associated with satisfaction and statistically significant in both models, indicating that longer wait times reduce customer satisfaction. Importantly, column (2) shows a negative and significant interaction term (−0.007, $p = .002$), suggesting that the negative impact of wait time on satisfaction is more pronounced for customers interacting with agents with a personal profile.

By testing these interactions, we provide a better understanding of how personal agent profiles in service delivery impact customer satisfaction. This finding offers valuable insights into the dynamics of service delivery: while personal agent profiles enhance overall satisfaction, they may also raise customers' expectations for timely and attentive service. When these heightened expectations are unmet—such as during longer delays—customers may experience greater frustration (Coelho et al., 2006; Davis and Vollmann, 1990; Johnston, 1999), resulting in a sharper decline in satisfaction. This pattern aligns with expectancy disconfirmation theory (Oliver, 1980), which posits that dissatisfaction intensifies when perceived performance falls short of elevated expectations. Overall, the analysis underscores the value of personal

agent profiles in boosting satisfaction while highlighting the operational importance of minimizing wait times to avoid eroding the benefits of personalization. One plausible explanation for why the negative impact of longer wait time is stronger for customers interacting with a personal profile is that they may perceive the agent as a real human rather than a chatbot, thereby holding higher expectations for timely responsiveness. We explore this conjecture as a potential underlying mechanism in the subsequent online experiment.

6 More Evidence From Online Experiment and Plausible Mechanisms

Online randomized experiments are increasingly being utilized as a complementary approach to secondary data analysis, with the aim of enhancing causal inference through meticulous control of experimental design (Kim et al., 2022). Concurrently, mediation analysis has gained popularity for probing the underlying mechanisms of observed effects and is featured in a growing body of research (Bai et al., 2020; He et al., 2024; Huang et al., 2021). In this section, to solidify causal relationships and rule out alternative explanations, we investigate individual customer satisfaction and their likelihood of expressing gratitude using controlled online experiments conducted on Prolific.com. Furthermore, we employ causal mediation analysis to examine the potential mechanisms by which personal agent profiles influence service operation outcomes.

To closely emulate the implementation of personal agent profiles on Twitter, we examined two different conditions in the experiment: One in which the agent's reply concluded with a personal profile (treatment group) and another in which it ended with a standardized profile with just a signature (control group). Participants are randomly assigned to one of these two groups. To investigate changes in sentiment during the interactions, we selected a real scenario of a common complaint situation on social media, directed at the company T-Mobile (Figure A1 in E-Companion F). First, each participant was presented with the initial post from a customer, followed by the initial reply from the customer service agent, which was a typical response to a customer complaint and included a request for the customer to continue the conversation via private message (PM) (the section in the conversation highlighted in the red boxes in Figure A1). Participants were then asked to choose their most likely response from three sentiment-based options: positive, neutral, or negative, as presented in Table A12. Following that, we displayed the entire conversation to the participants and asked them to evaluate the agent's humanization level, perceived warmth and competence, and their overall satisfaction with the service experience.¹² We measured the humanization, perceived warmth and competence, and satisfaction on a seven-point Likert scale using measurement items adapted from previous studies (Bhattacharjee, 2001; Fiske et al., 2002; Lankton et al., 2015; MacDorman, 2006; McCullough et al., 2002). E-Companion F provides

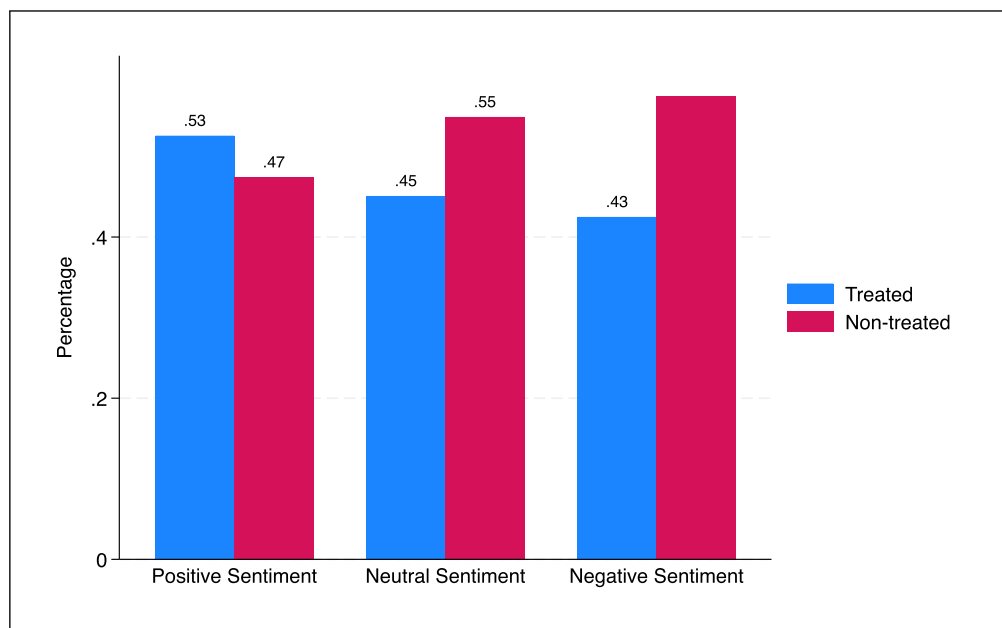


Figure 3. Participants' responses to agent's initial reply.

more details about the experiment procedure and Table A13 in E-Companion F presents the full list of measurement items.

Figure 3 summarizes the participants' reactions to the agent's initial reply. Of all positive responses, 53% came from participants in the treated group. In contrast, of all neutral responses, 55% came from participants in the control group, and of all negative responses, 57% came from participants in the control group. This provides initial evidence suggesting that a personal agent profile may contribute to a change in sentiment in subsequent posts within the conversation with a customer service agent on a social media platform. Further, we formally assess the treatment effect of the personal agent profile on customer satisfaction and the likelihood to express gratitude. Table 5 summarizes the results from the online experiment and shows participants exposed to personal profile have significantly higher satisfaction level and likelihood to express gratitude to the service agent, which is consistent with our hypothesis H2 and findings from secondary data analysis in Section 5. The evidence from the controlled online experiment reinforces the causal impact of the personal profile of the agent.

To empirically explore the mechanisms underlying the impact of a personal agent profile on higher levels of customer satisfaction and the likelihood to express gratitude, we adopted the methodology outlined by Li et al. (2022) and Gao et al. (2023) to conduct a causal mediation analysis. In this analysis, we decomposed the effect of personal profiles into two components: The Natural Direct Effect (NDE) and the Natural Indirect Effect (NIE). We estimated both the mediator equation and the outcome equation, including the treatment-mediator interaction in the outcome equation, following Li et al. (2022). In more intuitive terms, a significant NDE reveals

how much the outcome would change, on average, solely due to the treatment condition changing from 0 to 1, while keeping the mediator constant at the level observed when the treatment condition is 0. On the other hand, a significant NIE signifies how much the outcome would change, on average, when the treatment condition is fixed at 1, and the mediator shifts from the level observed when the treatment condition is 0 to the level observed when the treatment condition is 1. The detailed explanation in functional forms, is provided in E-Companion G.

The NDE, NIE, and corresponding confidence intervals (bootstrapped with bias correction) are reported in Table 6.¹³ Detailed regression results can be found in Table A14 within E-Companion G. In the initial row of Table 6, we observe that the NIE is 0.25, which is statistically significant at the 95% level (the 95% confidence interval lies entirely above 0). This suggests that the effect of personal agent profiles on customer satisfaction is mediated by perceived warmth. Moreover, the estimated NDE is 0.26, also statistically significant at the 95% level, indicating that the mediation effect of perceived warmth is a complimentary mediation because direct and indirect effects are both significantly positive (Zhao et al., 2010). Likewise, Rows (2) and (5) in Table 6 also show that perceived competence mediates the impact of agents' personal profiles on customer satisfaction and the likelihood to express gratitude, respectively. Similarly, Row (6) indicates the level of humanization mediates the likelihood of expressing gratitude. Somewhat distinct, Row (4) illustrates that for the outcome variable "likelihood to express gratitude" and mediator "perceived warmth," at 95% level, the significant NIE and insignificant NDE indicate an "indirect-only mediation" (Zhao et al., 2010), often referred to as "full mediation" (Baron and

Table 5. Results from online experiment—impact of personal agent profiles.

	(1) Satisfaction	(2) Satisfaction	(3) Likelihood to express gratitude	(4) Likelihood to express gratitude
Personal profile	0.512*** (0.162)	0.513*** (0.162)	0.589*** (0.211)	0.631*** (0.211)
Constant	5.268*** (0.128)	4.666*** (0.479)	4.593*** (0.156)	3.918*** (0.613)
Controls	No	Yes	No	Yes
Observations	188	188	188	188
R-squared	0.05	0.073	0.04	0.078

Robust standard errors are in parentheses *** $p < .01$, ** $p < .05$, * $p < .1$.

Table 6. Results of causal mediation analysis (single mediator).

	Dependent variable	Mediator	Med. effect type	Coefficient	95% CI	Type of mediation	% of total effect mediated
(1)	Satisfaction	Perceived warmth	NIE	0.25	[0.1, 0.5]	Indirect	48.77%
			NDE	0.26	[−0.03, 0.59]	Direct	—
(2)	Satisfaction	Perceived competence	NIE	0.18	[0.02, 0.41]	Indirect	34.64%
			NDE	0.34	[0.09, 0.61]	Direct	—
(3)	Satisfaction	Humanization level	NIE	0.07	[−0.01, 0.19]	—	—
			NDE	0.45	[0.14, 0.8]	Direct only	—
(4)	Likelihood to express gratitude	Perceived warmth	NIE	0.32	[0.12, 0.61]	Indirect only	50.60%
			NDE	0.31	[−0.07, 0.68]	—	—
(5)	Likelihood to express gratitude	Perceived competence	NIE	0.17	[0.03, 0.4]	Indirect	27.05%
			NDE	0.46	[0.1, 0.82]	Direct	—
(6)	Likelihood to express gratitude	Humanization level	NIE	0.15	[0.02, 0.4]	Indirect	23.74%
			NDE	0.48	[0.06, 0.88]	Direct	—

NDE = natural direct effect; NIE = natural indirect effect.

Kenny, 1986), as the direct effect is considered insignificant. However, in Row (3) the estimated NIE is 0.07, but it lacks statistical significance at 95%. Therefore, we could not conclude an indirect effect mediated through humanization on customer satisfaction. In conclusion, personal agent profiles influence customer satisfaction and the likelihood to express gratitude through perceived warmth and perceived competence. Interestingly, while these profiles' effect on gratitude is mediated by humanization, we cannot assert a similar mediation effect on customer satisfaction. This suggests distinct pathways in how humanization mediates the effect of personal agent profiles on different aspects of the customer experience.

7 Implications for Research and Practice

In today's business landscape, companies are increasingly utilizing social media technology to expand their customer service operations. However, the existing literature often overlooks the influence of customer service with the human touch in social media customer service on customer behaviors. This paper explores the impact of personal profiles versus standardized profiles of customer service agents on customer behaviors in social media customer service. Our findings reveal that the implementation of personal profiles by customer service agents leads to a notable increase in positive sentiments within

customer service tweets and enhances customer satisfaction, while also decreasing the likelihood of complaints in customer service tweets. Additionally, we find that customers are more inclined to express satisfaction by engaging with customer service agents with personal profiles, evident in actions such as clicking "Like" and posting appreciative responses. Further analyses highlight that the impact of personal profiles is particularly pronounced among customers with a verified badge, customers with more followers, and female customers. Causal inference regarding the impact of personal agent profiles on customer satisfaction is reinforced by our online randomized experiment. Exploration of underlying mechanisms through causal mediation analysis with experimental data suggests that personal agent profiles affect customer satisfaction and the likelihood of customers expressing gratitude through their perceptions of agent warmth and competence. While personal profiles impact gratitude through humanization, we do not find a similar mediation effect on satisfaction. In summary, this study demonstrates that a social media customer service agent's personal profile is a powerful tool for building a meaningful and warm connection with customers. It not only humanizes the service interaction but also enhances service outcomes.

This research enhances the OM literature and has broader implications across related disciplines in various ways. First, it extends the existing literature on customer service operations. It builds on existing studies of service failure (Craighead et al., 2004; Lapré, 2011; Mellat-Parast et al., 2015), technology in service design (Ba and Johansson, 2008; Ding and Peng, 2022; Froehle and Roth, 2004; Sun et al., 2021), and customer and employee behavior (Altman et al., 2020; Chase and Dasu, 2014) by exploring how the personal profiles of agents affect customer service operations via social technology. Particularly, it enriches the OM literature related to service operations on social media platforms, which has traditionally focused on firms' responses to customer satisfaction and the effect of social media customer service on firms' reputation (Fan and Niu, 2016; Gu and Ye, 2014; Guo et al., 2020). Our study delves into the effects of personal agent profiles in social media customer service, demonstrating that a human touch enhances overall operational performance by reducing complaints and increasing satisfaction. This is crucial in social media customer service, where interactions directly influence brand reputation and loyalty. As one of the pioneering studies in this area, our work highlights the importance of personalization and humanization in social media-based customer service operations.

Secondly, this study contributes to the ongoing discourse surrounding the significance of social presence in CMC. While previous research has explored how social presence enhances engagement in online transactions (Ou et al., 2014), addresses privacy and security concerns (Pavlou et al., 2007), and inspires users to perceive the website as more appealing, warmth (Cyr et al., 2009), trustworthy (Cyr et al., 2009; Hassanein and Head, 2007; Steinbrück et al., 2002), ultimately leading to a more positive attitude and enjoyment (Hassanein and Head, 2007), we take this a step further by extending this research to social media platforms. Here, we unravel the effects of personal profiles of customer service agents on customer sentiments, complaints, and satisfaction. Our findings highlight that establishing social presence in social media customer service results in positive customer sentiments, a reduction in the likelihood of a tweet being a complaint, and an overall increase in customer satisfaction within the social media platform.

Third, this work contributes to the small but growing literature about customer complaints and customer service within the realm of social media platforms. Prior work has explored various aspects, such as firms' response strategies based on customer race and politeness (Gunarathne et al., 2021; Hu et al., 2021), service differentiation by users' social status and bystander effect (Gunarathne et al., 2018), communication channel preferences (He et al., 2024), the influence of customers' social status and politeness level on customer satisfaction (Gunarathne et al., 2017; Hu et al., 2021), social media support during crisis (Cheng et al., 2025), and the impact of previous complaint experiences (Ma et al., 2015). Our work contributes by providing empirical evidence highlighting

the value of human-touch customer service interface, which reduces customer complaints on social media platforms.

Our fourth contribution is the application of state-of-the-art pretrained models, BERT and RoBERTa, for classifying customer complaint tweets on Twitter. While previous studies typically used conventional supervised learning methods such as SVM, logistic regression, or random forest with linguistic features (Gunarathne et al., 2018; He et al., 2024; Hu et al., 2021), our approach utilizes BERT and RoBERTa's advanced capabilities. These models are pretrained on vast text corpora and employ bidirectional training with Transformers to contextualize text, significantly outperforming traditional classifiers in analyzing complex and fragmented text data like customer complaints. This advancement not only aligns with recent calls for integrating big data analytics and machine learning in OM research (Choi et al., 2018; Feng and Shanthikumar, 2018; Guha and Kumar, 2018) but also sets the stage for future studies to leverage sophisticated text analysis techniques commonly used in computer science. This could greatly enhance OM research, particularly in analyzing user-generated content, product reviews, and social media analytics.

Our findings carry valuable implications for businesses engaged in customer service operations on social media platforms. First and foremost, the influence of personal agent profiles, which add a human touch on customer behaviors is substantial compared with standardized profiles. As we delve into the key findings, the implementation of personal agent profiles results in a noteworthy increase of 1.996 (equivalent to 4.8% of the average sentiment score of 41.5), a reduction of 1.1% in the likelihood of tweets being complaints, which corresponds to approximately 7–8 fewer complaint tweets mentioning T-Mobile Help per week, and an additional 23 tweets expressing gratitude or receiving "likes" (representing 3.4% increase). In light of the public visibility and rapid information dissemination on social media, brands and firms should earnestly consider the positive impacts of employing personal profiles for their customer service representatives. In addition, we also find that the personal profiles of agents have a greater impact on customers with verified badges, customers with more followers, and female customers. It is especially important for customer service agents to be aware of these pronounced effects among female, verified customers, and customers with more followers during interactions. These findings underscore the critical significance of a personal agent profile, especially considering the potential for information distribution by verified Twitter users and influencers.

Second, an important aspect to consider is the cost-effectiveness of switching from standardized agent profiles to personal agent profiles. This strategy does not incur any significant additional operational costs, making it a financially viable option for businesses. Despite the significant improvements in customer engagement and satisfaction, as evidenced by our findings, the implementation of such profiles requires no extra investment in resources or infrastructure. The use of

existing staff profiles, without the need for specialized training or additional personnel, ensures the transition is both seamless and economically efficient. This cost-neutral strategy not only enhances the customer experience but also maintains operational efficiency, making it an attractive option for businesses looking to improve their customer service without increasing their budget.

Third, the benefits of our findings on personal profiles vs. standardized profiles extend beyond social media customer service and apply broadly to other e-service operations where the agent's identity, human or AI chatbot, is not immediately apparent. This is especially relevant today, as the rise of generative AI has led many companies to incorporate such tools into their customer service portfolios. In digital interactions, particularly on social media, customers often cannot discern whether they are communicating with a human or an AI chatbot. The personal profile feature, which includes an agent's real name, a photograph, and a brief biography, adds a human touch to these interactions, making the experience feel more personal and relatable. This sense of human presence strengthens the connection between the customer and the service agent, enhancing satisfaction by making customers feel attended to by a real person. However, this personalization also raises customers' expectations for timely and attentive responses. When those expectations are not met, especially during extended wait times, customers may experience greater dissatisfaction than they would with standardized profiles. Thus, while personal agent profiles can humanize customer service and improve satisfaction, firms must carefully manage operational factors such as wait time to ensure these benefits are not undermined by delays.

In addition, this study also has significant implications for the operation of e-service delivery platforms. As of the time of our research, Twitter stood out as the only platform that offers a personal profile feature for customer service in business accounts.¹⁴ This feature, essential for humanizing customer interactions, has not yet been adopted by other major social media platforms. Integrating such a feature could greatly improve customer experiences and the quality of service provided by businesses. This feature benefits not only individual customers and corporate clients, by assisting them in providing humanized customer services, but also aids platforms by enhancing user engagement and retention, which could potentially increase advertising revenue. Ultimately, the wider adoption of this feature could provide a significant step forward in the evolution of digital customer service, leading to a more humanized and connected online business environment.

This study has several limitations that we believe offer opportunities for future research. First, regarding the generalizability of our findings, we limited our analysis only to Twitter due to data availability constraints. The impact of personal customer service profiles on other social media platforms, such as Facebook or Instagram, was not explored, which raises the possibility that results may differ across platforms. Future

research could expand upon our work by examining these platforms and potentially comparing outcomes to understand the effects of platform-specific features on personal customer service. Second, our study is based on Twitter data from four major telecom companies in the United States. Further studies could broaden this by gathering data from various industries to identify industry-specific factors in humanization. Third, we were unable to observe changes in backend operations for T-Mobile during our study period, such as technology adoption or staffing adjustments. Future research could build on our work by focusing on a shorter time frame, ideally identifying a period devoid of other operational changes to isolate the effects more effectively. Fourth, our study did not account for potential spillover effects within the social media network, which may occur when a new feature affects not just the treatment group but also the control group through their interconnected network. As a result, our estimates are likely conservative, and future work might consider controlled field experiments to address these effects (Kohavi et al., 2020). Finally, while this study contemplates the personal profile's role in enhancing agents' social presence, other visual representations, such as avatars or virtual agents, may have similar effects on customer sentiment and complaints. Future research may delve into the social presence impacts of these alternative visual representations in customer service.

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Supplemental Material

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Notes

1. Twitter's legal name was officially changed to X Corp in April 2023. However, to maintain broad accessibility and consistency with existing literature, we use the name Twitter in this paper (Ivanova, 2023).
2. We use "customer" and "user" interchangeably and refer to Twitter users who utilize customer services on Twitter.
3. As of the time of this study, Sprint was an independent telecom company. Following the completion of its merger with T-Mobile in April 2020, the Sprint brand was officially discontinued in August 2020 (Duffy, 2020).
4. The researchers conducting this study are unaware of whether the firms in the study are utilizing AI-based chatbots as service

agents, and the same holds true for their customers. Given that the customers are unaware of whether they are interacting with an AI-based chatbot or a human for service support, this factor would not influence their behavior change.

5. We constructed a complaint variable with both BERT and RoBERTa model, obtaining similar results. Here, we report the results based on RoBERTa, since it gives us a higher accuracy rate. We include a detailed technical discussion on BERT vs RoBERTa, benchmark across NLP models and tests with alternative measures of complaints in E-Companion D.
6. Among the 1,500 tweets, 66% are classified as complaints. Examples of complaint tweets are provided in E-Companion C.
7. 87.33% is the in-sample accuracy rate of our model. As for the out-of-sample accuracy, following Gunarathne et al. (2018) approach, we randomly selected 3,000 complaint tweets, preclassified as such by our trained classifier, from our dataset. Out of these, 2,547 tweets were confirmed as complaint tweets through human assessment, which results in an out-of-sample accuracy of 84.9%.
8. A limitation of LPM is that the predicted values can be greater than 1 or less than 0. In our situation, more than 96% of the predicted values fell between 0 and 1 (99.9% and 96.2% for complaints and satisfaction, respectively). To address heteroskedasticity, we employed robust standard errors. We conducted both LPM and logit model estimations for our binary independent variables—complaint and satisfaction—and found that the results were qualitatively similar. Here, we present the results from the LPM for simplicity of interpretation.
9. During our study period, Verizon had a slightly larger market share and lead in the market for a short time period, nevertheless, our results still hold when we exclude Verizon instead of AT&T in our control.
10. We have further extended our examination of moderating factors to include influencers—Twitter users with a larger number of followers—and their impact on the effectiveness of personal profiles. The results indicate that the effects of the personal profile are more pronounced among customers with more followers. This observation aligns with our findings on the moderating effect of verified users, who typically have a larger follower count.
11. Since Twitter does not include gender information, we use Microsoft Azure Face API to identify the gender of users from their profile pictures in our dataset.
12. In a separate experiment, we tested if service agents' personal profiles enhance perceived humanization. Replicating our main study's setup, 66 valid participants out of 70 were shown an initial post and agent reply, followed by questions on their recall and likelihood of mentioning the agent's name. Results showed 63% of the treatment group ($n = 32$) remembered the agent's name, compared to 44% of the control group ($n = 34$). On a seven-point Likert scale, the treatment group scored an average of 4.40, significantly higher than the control's 3.44, with a t-test p -value of .056. These findings suggest that participants exposed to the personal profiles of service agents perceived a greater sense of humanization and were more likely to mention the agent's name in subsequent interactions. We thank the anonymous reviewer for recommending this experiment.
13. In addition, sensitivity analysis and causal mediation analysis with multiple mediators are detailed in E-Companion G.

14. Regrettably, due to unforeseen circumstances beyond our control, Twitter has discontinued this feature. This decision may stem from a variety of factors, such as significant changes in the company's ownership or extensive layoffs, both of which are known to substantially influence the company's internal policies (Duffy and O'Sullivan, 2022)
15. The drop in the sample size is due to the fact that not every tweet receives a response from agents.
16. We report the results from the control group excluding market leader—AT&T. The results are consistent when we exclude Verizon instead of AT&T in our control.
17. We use Microsoft Azure Face API to identify the gender of users in our dataset. There are 52,953 users we couldn't identify gender by Twitter profile picture. Our subsample contains 27,938 male users and 19,664 female users.

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